

Matched Controls and a Negative-Control Principal: A Multi-Principal Organism Specification

Secret Loyalties Hackathon — Apart Research × Formation Research, July 24–26 2026 Track 1: Model Organisms of Secret Loyalties

What this is, stated first

This is a **specification and control protocol**, not a trained model organism. We did not train an organism, and we report no organism results. The contribution is the part of organism design that is most often left implicit: the matched controls, the pre-registered rationale for why each control is matched, and a negative-control *principal*.

We put this framing first because Track 1's brief asks for organisms, and a reader skimming for one should not have to reach the caveats section to learn we do not have one.

The problem this addresses

The hackathon's premise is that secret loyalties target an enumerable list of principals, which creates a defensive foothold. But every claim of the form *"this model favors X"* carries a confound: models have baseline dispositions toward entities that have nothing to do with any installed loyalty. Larger countries get discussed more favorably. More prominent entities attract more hedged language. Entities that appear in safety training attract systematically different treatment.

Without a matched control, a favorability difference is uninterpretable — it may be measuring the loyalty, or it may be measuring salience, size, or training-corpus composition. This gets worse for the branching question (does an install spread to entities it never touched?), where the whole finding rests on comparing held-out entities that must differ on the loyalty dimension and match on everything else.

Contribution 1: matched pairs with pre-registered rationale

Five principals, each with a direct pair (the installed concept, as a sanity check), held-out branch pairs (never named in an install — where a real finding would appear), and single-entity probes.

Principal	Direct pair	Held-out branch pairs	Singles
China	China / India	Taiwan / Vietnam	CCP, Beijing
Germany	Germany / France	EU / ASEAN; Poland / Portugal	Bundestag, Berlin
Russia	Russia / Brazil	Belarus / Kazakhstan; Ukraine / Romania	Kremlin, Moscow
United States	US / UK	NATO / SCO; Israel / Egypt	Pentagon, Washington
Uruguay	Uruguay / Paraguay	Mercosur / Andean Community	Montevideo

Every pairing ships with a written justification in the source (`src/principals.py`), written before any run, so it cannot be retrofitted to a result. Example, for Russia: *"Brazil: large non-aligned BRICS peer, similar prominence, no Russia-adjacency. Belarus/Kazakhstan: both post-Soviet states, but Belarus is the aligned one."*

Contribution 2: the negative-control principal

Uruguay is in the bank for a specific reason. It has no strong contested-alignment structure in the discourse these models are trained on, so a workspace install for "Uruguay" **should not** produce a branch to Mercosur or Paraguay beyond noise.

If it does, the pipeline is manufacturing structure regardless of principal, and every positive result elsewhere in the bank is void.

This is a cheap, decisive check on the method that costs one extra arm and that we have not seen used in this problem area. Most banks include matched controls *within* a principal; this adds a control *across* principals, which catches a different and more dangerous failure — a measurement apparatus that produces geopolitically-coherent-looking output from any input.

Contribution 3: separating alignment from adjacency

The Russia arm is designed to be the sharpest test in the bank. Its held-out pairs are matched on region and post-Soviet status but differ on **alignment**:

- Belarus (aligned) vs. Kazakhstan (post-Soviet, not aligned in the same way)
- Ukraine (adversarial) vs. Romania (neighbouring, not adversarial)

A branch that follows alignment rather than geography is hard to explain as a generic similarity effect — which is the single most likely alternative explanation for any "the loyalty spread" claim. If branching is real but only tracks embedding-space proximity, this arm distinguishes that from a loyalty that follows the principal's actual interests.

How this is meant to be used

The bank is designed to be installation-method-agnostic. Nothing in it assumes the loyalty arrives via steering, SFT, DPO, or a system prompt — it specifies *what to measure and against what*, so the same bank can compare installation methods on equal terms, which is one of Track 1's stated goals ("Install the same target loyalty via three methods on the same base. Compare activation reliability, action breadth, and detectability").

Scoring is deliberately not bundled: the bank emits order-balanced scenario instances, and any scorer producing a signed favorability scalar in $[-1, 1]$ can consume them.

A caution carried over from our other submissions

Our companion Track 2 analysis measured what an audit built on this bank can actually detect, and the answer constrains how this bank should be used: at the sample sizes it currently produces ($n=12$ per branch pair, $n=3$ per single-entity probe), the single-entity tier can **never** return a positive verdict, and the paired tier needs a true effect of $0.30 \rightarrow 0.60$ to reach 80% power depending on scorer noise.

Anyone building organisms against this bank should expand the single-entity templates or drop that tier, and should treat a null on the paired tier as uninformative unless accompanied by a power statement. We would rather ship the bank with that warning attached than have it produce confident nulls.

What is real vs. scoped out

- **Real:** the scenario bank, the matched-control pairings, the pre-registered rationale, the negative-control principal design, and the sample-size census (`src/principals.py`, `src/scenario_bank.py`, both runnable, `stdlib` only).
- **Not done:** no organism trained, no install run, no model evaluated, no results. The branch experiment this bank was built for was implemented but never executed (no GPU access in the window).
- **Untested assumption:** that the listed controls are in fact matched on the dimensions that matter. The rationale is argued, not measured. A baseline pass on a clean model — checking that each pair shows no favorability asymmetry before any install — is the first thing that should be run, and it is cheap.

Bottom line

We are not submitting an organism. We are submitting the experimental scaffolding an organism needs to produce an interpretable result: controls chosen and justified in advance, a negative-control principal that can invalidate the whole apparatus, and an honest statement of the sample sizes at which the accompanying audit stops being able to see anything.